Downscaling Municipal Soybean Yield Labels to Pixel Predictions with MSTNet in Paraná, Brazil

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Abstract

Official crop yield statistics are often reported at the level of administrative units, such as municipalities in Brazil and counties in the United States, rather than at finer spatial resolutions. However, precision agriculture, insurance, and localized risk assessment require estimates at finer spatial resolutions. We propose MSTNet (*Multi* STNet), a Transformer model that combines data sampled at different temporal frequencies. Separate STNet trunks encode soy-masked Sentinel-2 time series and a daily gridded climate series, while annual soil properties enter the decoder by late fusion. Since the model is able to predict a yield value per soy pixel, we can use it to predict yield for arbitrary polygons by averaging the pixel-wise predictions. Training uses the soy-masked municipality-mean Sentinel-2, climate, and soil features under PAM supervision. We also score every soy pixel after training and average those scores to municipalities, which tests how well the same weights transfer to arbitrary polygons. We evaluate in Paraná, Brazil, with year leave-one-out over harvest years 2020–2024, reporting only the left-out year in each fold. Excluding the 2022 drought fold, municipal-mean LOO R^2 is 0.275 and the pixel-path LOO R^2 is 0.302; when 2022 itself is left out, performance collapses because no comparable drought year remains in the training set.

1 Introduction

Reliable soybean productivity estimates matter for agricultural monitoring, insurance, and risk assessment. In Brazil, the Municipal Agricultural Production survey (PAM) from the Brazilian Institute of Geography and Statistics (IBGE) publishes official yields only at the municipality level. Similar reliance on aggregated statistics is found in other countries, including the United States (county-level) and India (district-level), where official yield estimates are commonly reported for administrative regions rather than individual fields. Sentinel-2 imagery, gridded weather, and soil maps describe crop condition at much finer scales, but field-level yield labels are rarely available statewide. The closest Brazilian precedent for learning soybean yield from IBGE municipal statistics and then mapping at pixel resolution is Song et al. [2022], who trained random forests on soy-masked Landsat/MODIS phenology metrics, climate, weather, topography, and soil, and produced annual 30 m maps for 2001–2020. Our goal is a model in that same municipal-to-pixel spirit: one that learns from municipal labels, predicts at smaller regions, and remains cheap enough to run over large areas.

MSTNet (*Multi* STNet) extends the Spectral–Temporal Network (STNet) of Xu et al. [2024]. Its Transformer encoder follows the self-attention architecture for satellite time series of Rußwurm and Körner [2020], and each covariate stream gets its own trunk when the streams are sampled at different frequencies. Here, one trunk encodes soy-masked Sentinel-2 reflectance and another encodes daily climate. Annual MapBiomas Solo properties, which are static once the harvest year is fixed, are concatenated in the decoder. Predictions are defined per soy pixel. Training fits the network under PAM supervision on soy-masked municipal-mean Sentinel-2 and climate series, with soil features fused in the decoder. The alternative would be to score every soy pixel during training and average before the loss; we avoid that path because it is much slower while targeting the same municipal labels. After training, we score every soy pixel of a given

municipality with the fitted weights and average those predictions for comparison with PAM, so the same model can also be aggregated to arbitrary polygons without retraining.

We run the experiments on soybean municipalities in Paraná for harvest years 2020–2024 under year leave-one-out: each fold trains on four years and scores only the left-out year. Because PAM is the only supervised target, every reported number is a municipal comparison with PAM, but under one of two input paths that we keep distinct throughout: the municipal-mean path used in training and model selection, and the post-training pixel path in which every soy pixel is scored and then averaged. Section 4 reports municipal-mean LOO metrics; Section 5 reports the pixel-path LOO metrics, maps, and scatterplots. Relative to Song et al. [2022], we replace hand-crafted phenology features and random forests by end-to-end multi-frequency Transformers on Sentinel-2 and daily climate sequences with late soil fusion. They evaluate mainly by holding out 20% of municipalities for testing while keeping every harvest year in both pools; we instead take year leave-one-out as the primary score and discuss that contrast in Section 6.1.

Our contributions are MSTNet itself, the pixel-scoring setup that turns municipal supervision into sub-municipal maps, and a five-fold year-LOO evaluation for Paraná over 2020–2024.

Section 2 describes the study area, PAM labels, and soy-masked covariates. Section 3 presents the MSTNet architecture, municipal-mean training and post-training pixel scoring, and the loss. Section 4 details the year-LOO design and municipal-mean LOO metrics, Section 5 reports the pixel-path results, and Section 6 covers the temporal versus spatial holdout contrast, the drought year, and limitations. Section 7 concludes. Appendix A diagrams the pixel-path data pipeline and architecture.

2 Data

The analysis uses municipal PAM productivity labels together with Sentinel-2, daily climate, and soil covariates (Table 1). All three covariate streams are gridded; for municipal aggregates we average them under the MapBiomas soybean mask. Figure 1 shows the municipality-mean path used for training and for the municipal-mean LOO metrics in Section 4. After training we also run a pixel-wise forward pass and average the scores for the pixel-path LOO metrics and sub-municipal maps in Section 5; Appendix A diagrams that pixel path.

Table 1. Summary of data sources used in the experiments.

Category	Variables	Source
Satellite	Ten Sentinel-2 L2A reflectance bands (B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12)	Copernicus Sentinel-2 L2A [Copernicus Sentinel-2, processed by ESA, 2021]; GEE [Gorelick et al., 2017]
Climate	Daily precipitation, ETo, Rs, Tmax, and Tmin	Xavier BR-DWGD [Xavier et al., 2022]
Soil	Clay, silt, sand, and soil organic-carbon stock at 0–30 cm	MapBiomas Solo Collection 3 [MapBiomas, 2025c,b]
Crop mask	MapBiomas soybean class 39, used to select cells before aggregating Sentinel-2, climate, and soil	MapBiomas Collection 10 [MapBiomas, 2025a]
Target	Municipal soybean productivity (t/ha)	IBGE PAM [Instituto Brasileiro de Geografia e Estatística, 2026]

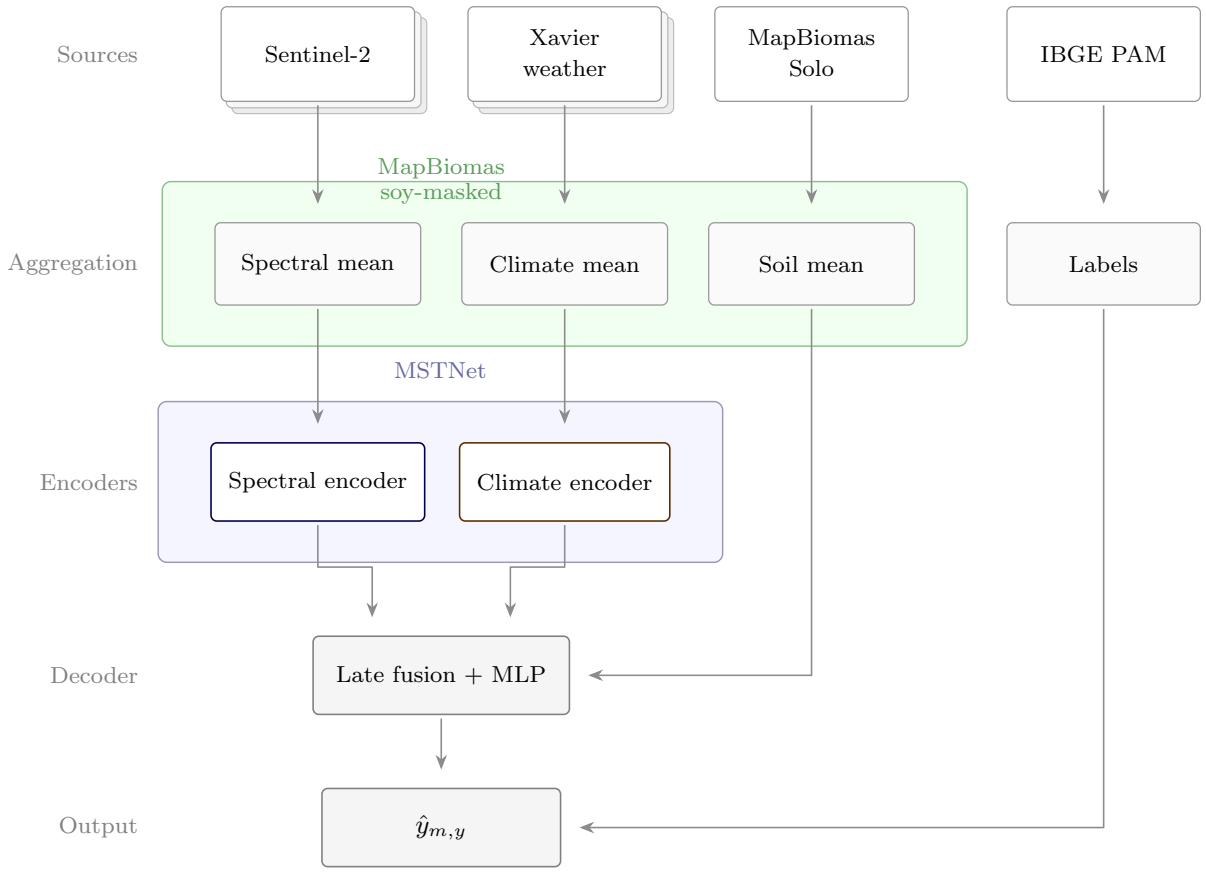

Figure 1. Data pipeline for MSTNet training (municipal-mean path). The MapBiomas soybean mask selects cells before municipal averaging of Sentinel-2, Xavier weather, and MapBiomas Solo; the spectral and climate means feed parallel STNet trunks, and soil enters the decoder by late fusion. The decoder prediction $\hat{y}_{m,y}$ is supervised by IBGE PAM labels at the output. The post-training pixel path is described in Section 3.2 and diagrammed in Appendix A.

2.1 Study area and labels

The study covers soybean-producing municipalities in the state of Paraná, Brazil. The target is municipal soybean productivity in tonnes per hectare (t/ha) from IBGE PAM [Instituto Brasileiro de Geografia e Estatística, 2026]; municipality code and harvest year identify each observation. No field- or pixel-level yield labels are used. For each municipality–year we compare the MapBiomas soybean area with the PAM planted area and keep only pairs whose ratio lies in $[0.5, 1.5]$. This drops cases where the mask covers far less or far more soy than PAM reports, so the soy-masked covariates are not badly mismatched to the official area.

2.2 Covariates and municipal aggregation

Both Sentinel-2 imagery and daily climate covariates cover the same inclusive Oct 1–Mar 31 window. We use this fixed window as an approximation of the soybean growing season in Paraná, assuming sowing begins around Oct 1 and harvest is completed by Mar 31. Sentinel-2 L2A reflectance is processed in Google Earth Engine [Gorelick et al., 2017]: for each municipality and observation date we compute soy-masked zonal means of ten bands (B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12), using MapBiomas Collection 10 class 39 as the soybean mask [MapBiomas, 2025a]. Each municipal spectral sequence retains a day-of-year index, and the longest observed sequence contains 45 observations. Climate covariates come from the Xavier daily weather

grid [Xavier et al., 2022]: precipitation, reference evapotranspiration (ETo), solar radiation (Rs), maximum temperature (Tmax), and minimum temperature (Tmin). The window contains 182 daily observations in ordinary seasons and 183 when February has 29 days. As with reflectance, daily climate is averaged under the MapBiomas soybean mask and scaled independently of Sentinel-2; MSTNet encodes it in a dedicated trunk rather than concatenating climate onto Sentinel-2 timesteps. For each municipality–season, we construct a four-value MapBiomas Solo soil vector comprising static clay, silt, and sand fractions [MapBiomas, 2025c] and the annual SOC stock for the corresponding harvest year [MapBiomas, 2025b], all at 0–30 cm. These properties are likewise soy-masked, and the resulting per-season vector is late-fused in the decoder.

2.3 Year leave-one-out splits

Evaluation follows year leave-one-out over harvest years 2020–2024 (Section 4.2). In each fold, every municipality–year from the four training years is used for fitting; the left-out year is never seen in training and provides the only reported municipal scores for that fold. Municipal-mean LOO metrics appear in Section 4; pixel-path LOO metrics and maps appear in Section 5. We do not validate the pixel-level maps against field-level yields because no field-level ground-truth data are available.

3 Methodology

We first describe the MSTNet architecture (STNet trunks and late-fusion head), then how the network is trained under municipal PAM labels and scored at pixel level after training, and finally the loss.

3.1 MSTNet architecture

MSTNet (*Multi* STNet) places one STNet trunk on each time-varying stream so that modalities with different sampling rates need not share a common calendar. The single-trunk STNet material and notation below follow Xu et al. [2024]; their encoder builds on the self-attention architecture for satellite time series of Rußwurm and Körner [2020]. We add trunk indices s and c for the dual-stream case and replace the classification head by a regression decoder with late soil fusion. Figure 2 summarizes the network.

Sequential encoding. Write a satellite imagery time series (SITS) as $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_T]$ with $\mathbf{x}_t \in \mathbb{R}^C$. In MSTNet the spectral trunk takes $\mathbf{X}_s$ with $C_s = 10$ Sentinel-2 bands and length $T_s = 45$, the number of clear observations in the municipality with the densest Sentinel-2 series over the Oct–Mar window. Shorter municipal series are zero-padded and masked to this length. The climate trunk takes $\mathbf{X}_c$ with $C_c = 5$ climate channels over the Oct–Mar window. That window has 183 days in leap years, when February has 29 days, and 182 days otherwise, so we fix $T_c = 183$ and pad ordinary seasons by one masked step. Each timestep is mapped by a trunk-specific, weight-sharing MLP,

$$\mathbf{e}_{i,t} = MLP_i(\mathbf{x}_{i,t}), \tag{1}$$

and the trunk embedding matrix stacks these vectors as

$$\mathbf{E}_i = [\mathbf{e}_{i,1}, \ldots, \mathbf{e}_{i,T_i}] \in \mathbb{R}^{T_i \times d}, \tag{2}$$

where $i \in \{s, c\}$. Because self-attention is permutation-equivariant, a day-of-year (DoY) positional vector $\mathbf{p}_{i,t} \in \mathbb{R}^d$ is added at every step,

$$\mathbf{e}'_{i,t} = \mathbf{e}_{i,t} + \mathbf{p}_{i,t}, \tag{3}$$

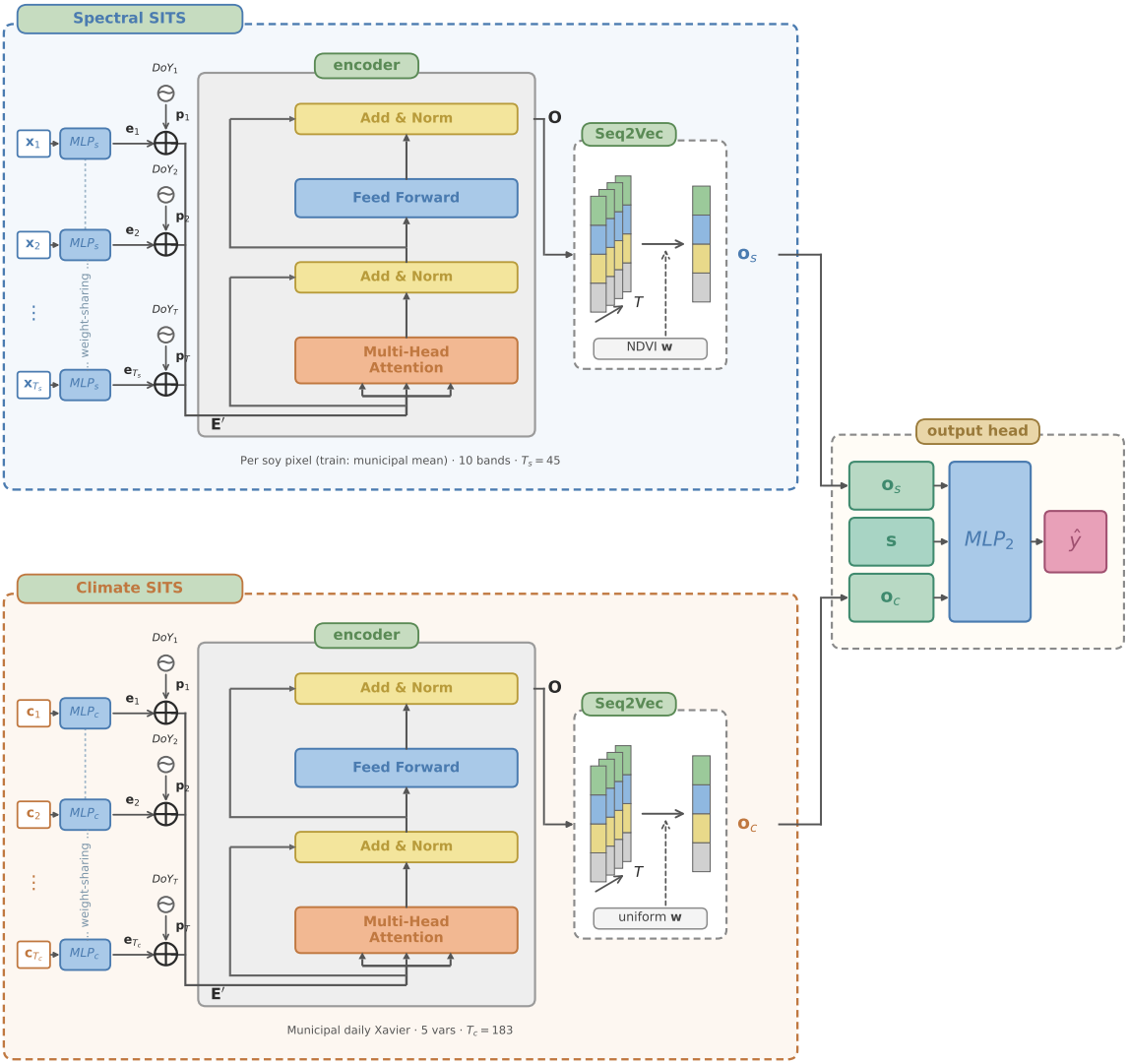

Figure 2. MSTNet architecture. Each STNet trunk $i \in \{s, c\}$ maps a SITS $\mathbf{X}_i$ of length T_i through a shared per-timestep MLP to $\mathbf{E}_i$, adds DoY encodings to form $\mathbf{E}'_i$, applies a Transformer encoder to obtain $\mathbf{O}_i$, and pools with Seq2Vec to $\mathbf{o}_s$ with NDVI weights or $\mathbf{o}_c$ as a mean over valid days. The output head concatenates $\mathbf{o}_s$, $\mathbf{o}_c$, and MapBiomas Solo features $\mathbf{s}$, then MLP_2 predicts productivity $\hat{y}$. Adapted from Fig. 2 of Xu et al. [2024].

with

$$\mathbf{p}_{i,t} = \big[\sin(\mathrm{DoY}_{i,t}/\tau^{2k/d}),\, \cos(\mathrm{DoY}_{i,t}/\tau^{2k/d})\big]_{k=1}^{d/2}, \tag{4}$$

and $\tau = 1000$, which gives $\mathbf{E}'_i = [\mathbf{e}'_{i,1}, \ldots, \mathbf{e}'_{i,T_i}]$. The DoY calendars differ by trunk: spectral steps follow the clear Sentinel-2 acquisition dates, while climate steps follow the daily Oct–Mar grid.

Transformer encoder. Each trunk applies a Transformer encoder to $\mathbf{E}'_i$. Inside each encoder layer, multi-head self-attention forms query, key, and value projections at every timestep, runs scaled dot-product attention over the sequence in parallel heads, and recombines the head outputs. A position-wise feed-forward network follows, and both sublayers use residual connections with layer normalization, yielding $\mathbf{O}_i = [\mathbf{o}_{i,1}, \ldots, \mathbf{o}_{i,T_i}] \in \mathbb{R}^{T_i \times d}$.

Seq2Vec. A Seq2Vec module collapses $\mathbf{O}_i$ to a single vector $\mathbf{o}_i \in \mathbb{R}^d$. On the spectral trunk we use an NDVI-weighted temporal mean: with NIR and red channels of $\mathbf{x}_{s,t}$,

$$\mathrm{NDVI}_t = \frac{x_{s,t}^{\mathrm{NIR}} - x_{s,t}^{\mathrm{Red}}}{x_{s,t}^{\mathrm{NIR}} + x_{s,t}^{\mathrm{Red}}}, \tag{5}$$

$$\mathbf{w} = \mathrm{softmax}\big([\mathrm{NDVI}_t]_{t=1}^{T_s}\big), \tag{6}$$

$$\mathbf{o}_s = \sum_{t=1}^{T_s} w_t\, \mathbf{o}_{s,t}, \tag{7}$$

so cloudy or low-vegetation dates receive small weight. On the climate trunk there is no NDVI analogue; we take the mean of $\mathbf{O}_c$ over valid (unmasked) days, denoted $\mathbf{o}_c$.

Output head. Since MapBiomas Solo properties $\mathbf{s}$ are static once the harvest year is known, we fuse them directly in the decoder. MSTNet concatenates the trunk embeddings with soil and decodes a scalar productivity estimate,

$$\hat{y} = MLP_2\big([\mathbf{o}_s; \mathbf{o}_c; \mathbf{s}]\big), \tag{8}$$

with $\hat{y}$ interpreted per pixel or per municipal aggregate according to the input path in Section 3.2.

3.2 Municipal-mean training and post-training pixel scoring

Let $\mathcal{P}_{m,y}$ denote the soybean pixels observed in municipality m in harvest year y. Training uses the municipal-mean path: every covariate stream is restricted to those soy pixels and averaged before it enters the network. For Sentinel-2 reflectance at date t,

$$\bar{\mathbf{x}}_{m,y,t} = \frac{1}{|\mathcal{P}_{m,y,t}|} \sum_{p \in \mathcal{P}_{m,y,t}} \mathbf{x}_{p,t}, \tag{9}$$

and the same soy-masked average yields the municipal climate series $\bar{\mathbf{c}}_{m,y}$ and soil vector $\bar{\mathbf{s}}_{m,y}$. Each municipality–year is then one forward pass $\hat{y}^{\mathrm{train}}_{m,y} = f_\theta(\bar{\mathbf{x}}_{m,y}, \bar{\mathbf{c}}_{m,y}, \bar{\mathbf{s}}_{m,y})$ supervised by PAM (Figure 1).

After training, the same weights define a productivity score $\hat{y}_p$ for each soy pixel p from its Sentinel-2 sequence $\mathbf{x}_p$, the municipal daily climate series $\mathbf{c}_{m,y}$ (broadcast to all pixels in m), and soil covariates $\mathbf{s}_p$. Each pixel score is

$$\hat{y}_p = f_\theta(\mathbf{x}_p, \mathbf{c}_{m,y}, \mathbf{s}_p), \tag{10}$$

and municipal predictions average those scores,

$$\hat{y}_{m,y} = \frac{1}{|\mathcal{P}_{m,y}|} \sum_{p \in \mathcal{P}_{m,y}} \hat{y}_p, \tag{11}$$

then compared with the PAM label $y_{m,y}$. The same field $\{\hat{y}_p\}$ can be kept as a map or averaged over any sub-municipal polygon. This post-training pixel path is the source of the LOO metrics, maps, and scatterplots in Section 5. Appendix A diagrams the corresponding data pipeline and recalls the shared MSTNet architecture at pixel scoring time. The same forward path would be used to train pixel-wise by scoring every soy pixel and averaging before the loss; we do not train that way because it is much slower while targeting the same municipal labels. Because f_θ is nonlinear, mean-then-encode and encode-then-mean need not coincide. Section 4.3 reports the same LOO years under the municipal-mean path.

3.3 Loss

Let $y_{m,y}$ be the PAM productivity for municipality m in harvest year y, and let μ and σ be the mean and standard deviation of PAM targets on the training years of the current fold. Targets are standardized as

$$\tilde{y}_{m,y} = \frac{y_{m,y} - \mu}{\sigma}, \tag{12}$$

and the municipal-mean forward pass outputs a prediction $\hat{y}^{\text{train}}_{m,y}$ in the same standardized space. Training minimizes the mean squared error over the municipal-mean training set $\mathcal{D}$,

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{(m,y) \in \mathcal{D}} \left(\hat{y}^{\text{train}}_{m,y} - \tilde{y}_{m,y}\right)^2. \tag{13}$$

Within each leave-one-year-out fold, μ, σ, and the channel-wise standardization statistics for Sentinel-2, climate, and soil are recomputed using only the four training years and then applied unchanged to the held-out year. Reported metrics (RMSE, MAE, and R^2) use the inverse map $\sigma\hat{y} + \mu$ back to t/ha.

4 Experiments

We first fix the MSTNet training setup, then define the year leave-one-out design, report municipal-mean LOO metrics, and describe the leave-one-modality ablations. Section 5 presents the corresponding pixel-path scores, maps, and ablation tables.

4.1 Proposed model

The proposed model is MSTNet: one STNet trunk for the Sentinel-2 series, one for the daily climate series, and late fusion of soil features in the decoder. Each trunk uses one Transformer encoder layer with $d = 128$, four attention heads, feed-forward width $d_{\text{inner}} = 64$, and dropout 0.4. Optimization uses AdamW with learning rate 1×10^{-3}, weight decay 5×10^{-4}, and batches of 40 municipalities. The learning rate drops at epochs 40 and 120; training stops early after fifteen epochs without improvement in the validation loss. Training and checkpoint selection use the municipal-mean path (Section 3.2).

4.2 Year leave-one-out design

We evaluate temporal generalization with year leave-one-out cross-validation (LOO) over harvest years 2020–2024. For each fold with left-out year Y, all municipality–years with harvest year $\neq Y$ are used for training on the municipal-mean path, and year Y is held out entirely so that

no municipality from Y enters training. Model selection within the fold uses the municipal-mean validation loss on year Y; after training, each fold is scored on year Y under both the municipal-mean path and the post-training pixel path. Input standardization is refit on each fold's training years. This yields five independently trained checkpoints, one per left-out year.

4.3 Municipal-mean LOO metrics

Table 2 reports LOO metrics when the same fold checkpoints are evaluated on the municipal-mean path used in training. Across the five folds, mean R^2 is 0.160 and mean RMSE is 0.617 t/ha. Excluding the 2022 drought fold, mean R^2 is 0.275 and mean RMSE is 0.435 t/ha. On non-drought folds, R^2 ranges from 0.199 in 2023 to 0.410 in 2021, the strongest year, with RMSE between 0.320 and 0.577 t/ha. The 2022 fold collapses to $R^2 = -0.300$ and RMSE = 1.342 t/ha, so the five-fold mean is pulled down by that single out-of-distribution season. Section 5 reports the corresponding post-training pixel-path scores, maps, and scatterplots.

Table 2. Year leave-one-out metrics for MSTNet under the municipal-mean evaluation path (mean-then-encode). Each row evaluates only the listed LOO year with the fold trained on the other four harvest years. RMSE and MAE are in t/ha.

LOO year	n	RMSE	MAE	R^2
2020	346	0.411	0.316	0.233
2021	343	0.320	0.239	0.410
2022	342	1.342	1.069	−0.300
2023	339	0.433	0.345	0.199
2024	336	0.577	0.445	0.259
Mean	—	0.617	0.483	0.160
Mean excl. 2022	—	0.435	0.336	0.275

4.4 Pixel-path evaluation

For each LOO fold we also compare municipal predictions with PAM labels on the held-out year under the post-training pixel path (11). The same pixel path is re-run with truncated seasons that keep only the first k months of the Oct–Mar window for $k = 1, \ldots, 6$, so that early-season R^2 can be compared with the full-window score. Section 5 reports the full-season per-fold RMSE, MAE, and R^2 values, the partial-season R^2 curves, predicted-versus-observed scatterplots for every LOO year, municipal error and Mamboré maps for the best and worst folds, and a 2022 Mamboré comparison against mean and peak NDVI. Appendix B reports the municipal-mean counterparts of the partial-season curves, scatterplots, and Paraná error maps; Appendix D adds the remaining pixel-path heatmaps.

4.5 Modality ablations

To isolate each modality we retrain three ablations on the same year-LOO folds as the full model, each omitting one input stream: no soil (Sentinel-2 + climate), no climate (Sentinel-2 + soil), and no Sentinel-2 (climate + soil). Hyperparameters match full MSTNet; only the inputs change—late soil fusion is dropped, or the climate trunk is removed, or the Sentinel-2 trunk is removed. Checkpoint selection still uses municipal-mean validation loss on the held-out year; Section 5.5 and Appendix B.3 report pixel-path and municipal-mean scores.

5 Results

We report post-training pixel-path LOO accuracy against PAM, then examine the 2022 drought fold, partial-season truncations, scatterplots and spatial maps, and modality ablations. Municipal-mean LOO metrics for the same checkpoints are in Section 4.3; the corresponding municipal-mean scatterplots, error maps, and partial-season curves are in Appendix B.

5.1 Year-LOO municipal accuracy

Table 3 reports post-training pixel-path metrics on the left-out year of each fold: every soy pixel is scored and the scores are averaged to the municipality before comparison with PAM. Deltas are relative to the municipal-mean path on the same checkpoints (Table 2). Across the five folds, mean R^2 is 0.157 and mean RMSE is 0.623 t/ha. Excluding the 2022 drought fold, mean R^2 rises to 0.302 and mean RMSE falls to 0.427 t/ha, a mean ΔR^2 of +0.027 over the municipal-mean path. The strongest non-drought fold is 2021 ($R^2 = 0.402$, RMSE = 0.323 t/ha).

Table 3. Year leave-one-out metrics for MSTNet (pixel-by-pixel scores averaged to municipalities). Each row is a fold that trains on the other four harvest years and evaluates only the listed LOO year. RMSE, MAE, ΔRMSE, and ΔMAE are in t/ha. Each Δ is the pixel-path metric minus the municipal-mean-path metric from Table 2 on the same fold; negative ΔRMSE or ΔMAE and positive ΔR^2 mean the pixel path improves on the municipal-mean path.

LOO year	n	RMSE	MAE	R^2	ΔRMSE	ΔMAE	ΔR^2
2020	346	0.395	0.316	0.291	−0.016	0.000	+0.058
2021	343	0.323	0.244	0.402	+0.003	+0.005	−0.008
2022	342	1.404	1.136	−0.422	+0.062	+0.067	−0.122
2023	339	0.424	0.337	0.231	−0.009	−0.008	+0.032
2024	336	0.568	0.439	0.284	−0.009	−0.006	+0.025
Mean	—	0.623	0.495	0.157	+0.006	+0.012	−0.003
Mean excl. 2022	—	0.427	0.334	0.302	−0.008	−0.002	+0.027

5.2 Drought year 2022

Harvest 2022 is the only year in our 2020–2024 sample with a severe drought across Paraná. When 2022 is the LOO year, the training set contains no analogous drought season, so the model must extrapolate outside the climatic regimes seen in training. That fold collapses: $R^2 = -0.422$ and RMSE = 1.404 t/ha, far worse than the other LOO years. Figure 5 places that drought error map beside the best non-drought fold, 2021. The 2022 panel shows a spatially coherent band of large positive errors across the west and northwest of the state: the model over-predicts where observed PAM productivity collapsed. That footprint matches the known geography of the 2021/22 drought, which hit the West and Northwest hardest, including the Toledo, Cascavel, and Paranavaí regions. Municipalities such as Palotina, Terra Roxa, Medianeira, Toledo, and Assis Chateaubriand appear among the largest over-predictions in our LOO fold. The remaining folds, trained with 2022 available when another year is held out, do not face that missing-regime problem to the same degree. We therefore treat 2022 as a stress test of out-of-distribution seasons rather than as a typical LOO score, and we report means both with and without that fold.

5.3 Partial-season evaluation

After training on the full Oct–Mar window, we also score each LOO fold on truncated inputs that keep only the first k season months, for $k = 1, \ldots, 6$, on the pixel path. Table 4 and Figure 3 report municipal R^2 as months are added. With one or two months the scores are strongly negative on every fold. Positive R^2 appears around $k = 4$ on non-drought years and approaches the full-season value by $k = 5$–6: excluding 2022, mean R^2 rises from -0.156 at three months to 0.165, 0.288, and 0.302 at four, five, and six months. The small gain from adding March, relative to stopping in February, suggests that on the observed years either harvest is largely complete before March begins or almost all information used by the model is already available by late February; we cannot separate these explanations because the actual harvest dates are not observed. The drought fold stays negative for every truncation. Appendix B reports the same truncations under the municipal-mean path.

Table 4. Pixel-path year-LOO R^2 under incomplete seasons. Each column keeps only the first k months of the Oct–Mar window ($k = 1$ is October only; $k = 6$ is the full season reported in Table 3).

LOO year	Oct	Nov	Dec	Jan	Feb	Mar
2020	−9.763	−0.302	0.050	0.162	0.257	0.291
2021	−7.178	−8.708	−0.752	−0.013	0.387	0.402
2022	−1.049	−0.931	−0.692	−0.500	−0.443	−0.422
2023	−2.434	−2.326	−0.263	0.200	0.238	0.231
2024	−2.754	−2.407	0.343	0.312	0.272	0.284
Mean	−4.636	−2.935	−0.263	0.032	0.142	0.157
Mean excl. 2022	−5.532	−3.436	−0.156	0.165	0.288	0.302

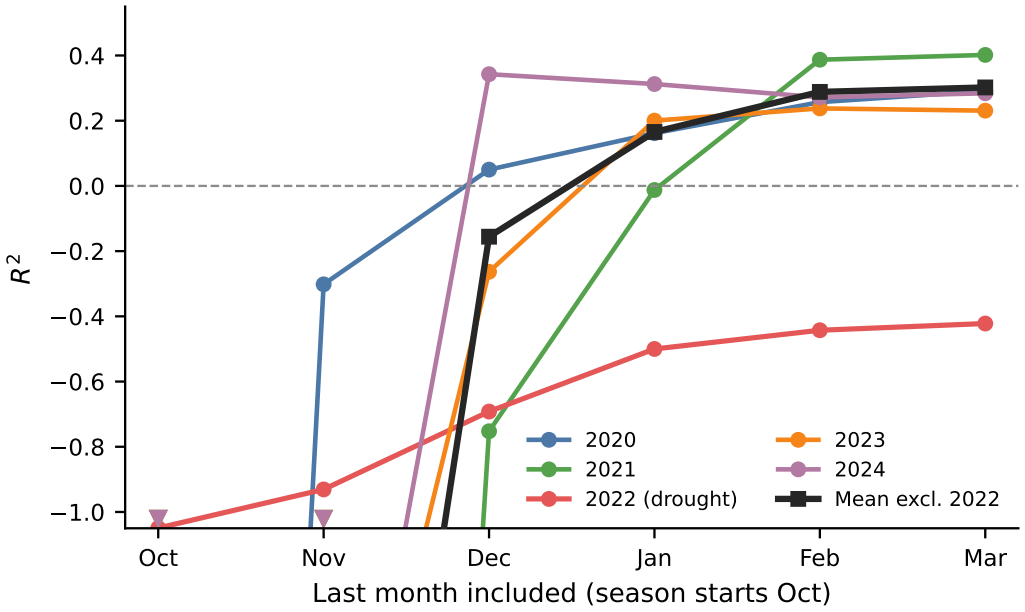

Figure 3. Pixel-path LOO R^2 versus season length. Each curve is one left-out year; the black squares are the mean over non-drought folds. Points below the axis floor are clipped and marked with downward triangles.

5.4 Scatterplots and spatial examples

Figure 4 shows predicted versus observed municipal productivity for every LOO year. Non-drought folds track the identity line with moderate scatter; the 2022 panel shows the systematic failure under drought. Even on better years, predictions compress toward a mid-range value: many municipalities are scored near the same average productivity, and the model under-separates the highest-yielding municipalities from the rest.

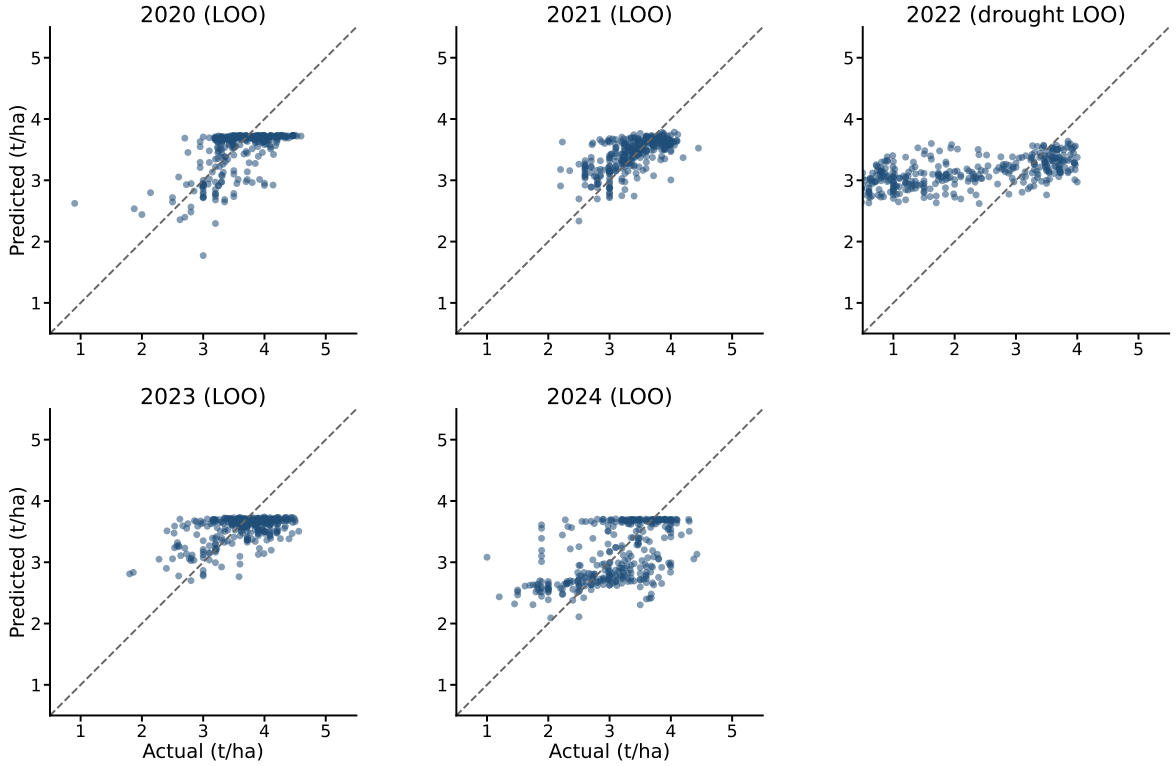

Figure 4. Predicted versus observed municipal soybean productivity (t/ha) for each year-LOO fold of MSTNet under the pixel path. Every panel evaluates only the left-out year. The dashed line is identity.

Figure 5 maps municipal aggregate errors for the best and worst LOO years on the pixel path, 2021 and 2022. Relative to the more scattered 2021 residuals, the 2022 errors concentrate where the drought hit hardest, so the spatial pattern of model failure is itself informative about the missed regime. The remaining years appear in Appendix D.

Figure 6 shows pixel-level scores for Mamboré under those same two checkpoints. This is the main goal of the method: after training only on municipal PAM labels, the network can score arbitrary polygons, here the soy pixels inside one municipality, without field-level yields. Those maps remain qualitative; we have no field yields to score them against, only the municipal PAM total when the pixels are averaged. Appendix D adds the remaining Mamboré years and a stacked comparison of the 2022 productivity map with mean and peak NDVI.

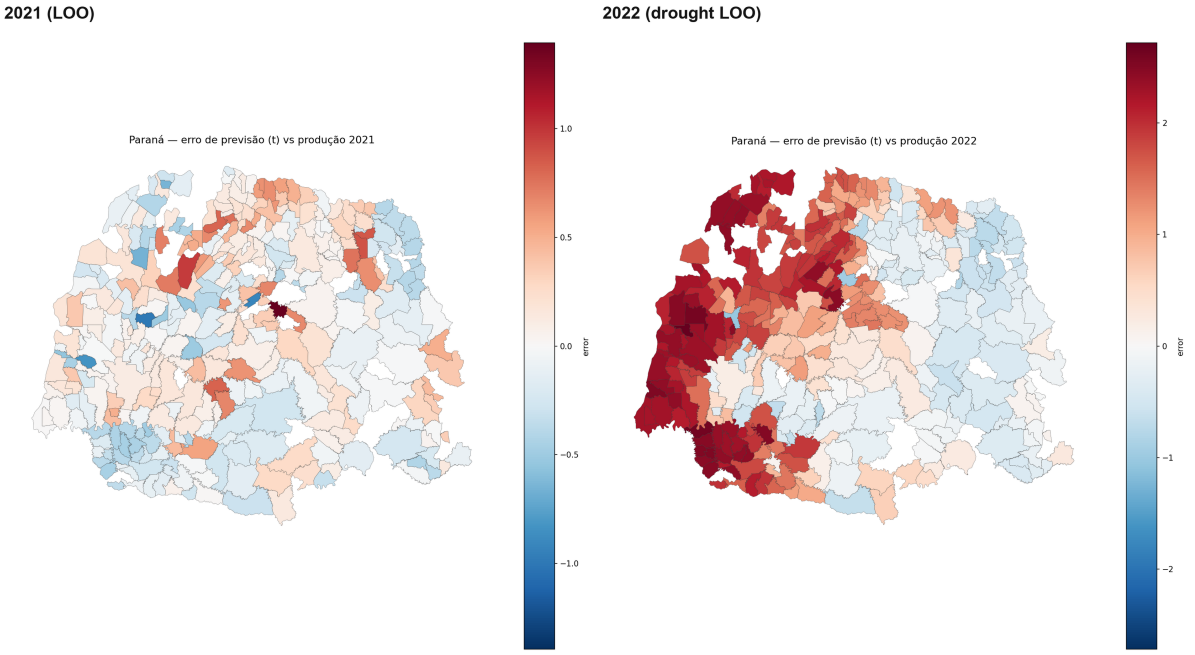

Figure 5. Municipal aggregate prediction error maps under the pixel path for the best non-drought LOO year (2021) and the drought LOO year (2022).

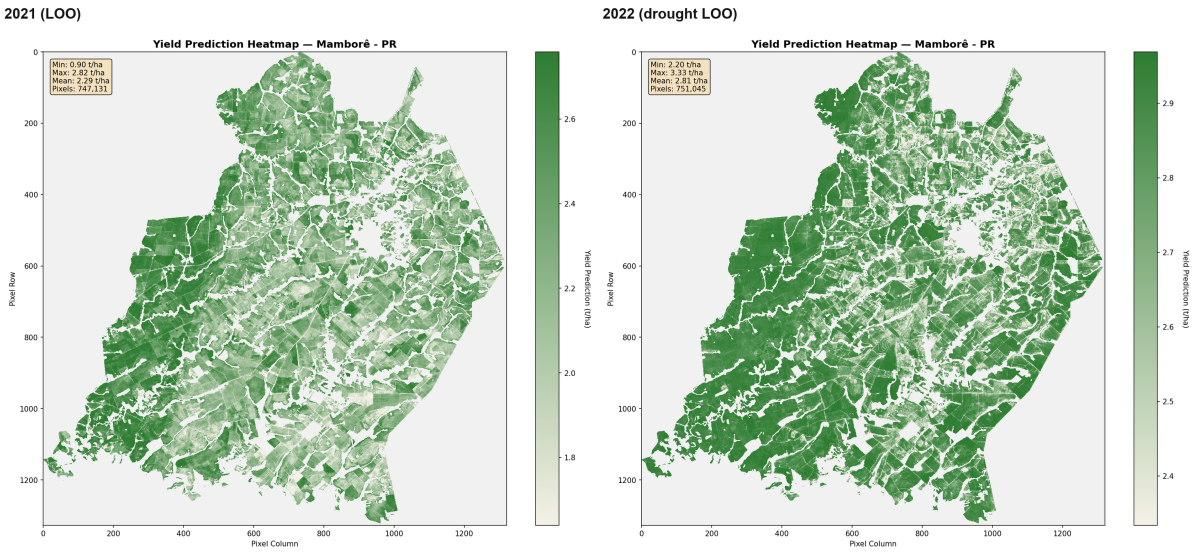

Figure 6. Intramunicipal maps of downscaled pixel-level productivity predictions for Mamboré, Paraná, under the 2021 and 2022 LOO checkpoints.

5.5 Ablations

Table 5 reports the fixed-architecture leave-one-modality runs from Section 4.5 on the pixel path; Appendix B.3 reports the municipal-mean counterparts. Excluding 2022, full MSTNet has mean $R^2 = 0.302$ and RMSE $= 0.427$ t/ha. Omitting climate costs about 0.32 in mean R^2, omitting Sentinel-2 about 0.26, and omitting soil about 0.07. On the 2022 fold, removing soil lifts R^2 from -0.422 to -0.203, but non-drought years still favor the full three-stream setup, so the five-fold mean does not rise above the full model.

Table 5. Year-LOO pixel-path metrics for full MSTNet and leave-one-modality-out ablations. Each ablation uses the same hyperparameters as full MSTNet and drops only one input stream, then scores soy pixels and averages to municipalities. RMSE is in t/ha. Full MSTNet keeps S2+climate+soil; No soil keeps S2+climate; No climate keeps S2+soil; No Sentinel-2 keeps climate+soil.

Variant	Metric	2020	2021	2022	2023	2024	Mean	Excl. 2022
Full MSTNet	R^2	0.291	0.402	−0.422	0.231	0.284	0.157	0.302
	RMSE	0.395	0.323	1.404	0.424	0.568	0.623	0.427
No soil	R^2	0.232	0.317	−0.203	0.115	0.273	0.147	0.234
	RMSE	0.412	0.345	1.291	0.455	0.572	0.615	0.446
No climate	R^2	0.163	−0.109	−0.491	−0.203	0.081	−0.112	−0.017
	RMSE	0.430	0.439	1.438	0.531	0.643	0.696	0.511
No Sentinel-2	R^2	−0.438	0.197	−0.297	0.105	0.298	−0.027	0.040
	RMSE	0.563	0.374	1.341	0.458	0.562	0.659	0.489

6 Discussion

We discuss how MSTNet relates to Song et al. [2022], how mean-then-encode training relates to encode-then-mean pixel evaluation, what the 2022 drought fold implies for operations, and where the present features, architecture, and losses may still fall short, with a few ideas for future work.

6.1 Temporal versus spatial holdout

To the best of our knowledge, the closest counterpart in aim is Song et al. [2022]: they use Brazilian soybean yield at municipal scale from multi-year satellite, climate, and soil inputs under IBGE PAM labels, while keeping the model capable of predicting at the pixel level. In the main results we focused on the temporal holdout. As Song et al. [2022] also did, however, a spatial holdout is another valid way to measure what the model can do. The two ask different questions. A spatial split that holds out part of the (municipality, year) pairs tests whether the model can interpolate to unseen places under seasons already observed in training, which is useful for downscaling and gap-filling within known years. Leave-one-year-out keeps the places and drops a calendar year, testing whether the model can generalize to an unseen season, closer to forecasting the current year. Temporal holdout is typically harder because the model cannot rely on having already seen that year's climate regime, phenology, or yield level among the training municipality–years; spatial holdout can still borrow seasonal structure shared across the panel.

Very briefly on the numbers: Song et al. [2022] report spatial-holdout $R^2 = 0.69$ (RMSE 344 kg/ha) and leave-one-year-out RMSE from 259 to 816 kg/ha, with a mean per-year leave-one-year-out R^2 of about 0.42 across 2001–2019. Under the same spatial split on Paraná, MSTNet reaches pooled $R^2 = 0.736$ (RMSE 0.448 t/ha; Table 8), with yearly holdout R^2 from 0.283 to

0.636 (mean 0.467). That puts our spatial performance roughly in line with theirs, at least on the spatial split. The pooled score is also higher than the best single-year R^2 (0.636), which is the signature of strong between-year separation: when all harvest years sit in one pool, getting the seasonal level right across wet and dry years inflates R^2, whereas within a fixed year the model has less room to separate municipalities. A caveat is the panel: we have only five harvest years in Paraná, whereas they observe nineteen years across many Brazilian states. More years are clearly helpful for temporal coverage. More states can help if neighbouring regions share yield-relevant structure, but can also add geographic and management complexity that a single model must absorb.

6.2 Mean-then-encode versus encode-then-mean

Municipal-mean evaluation and the post-training pixel path are not the same estimator. The former applies f_θ to soy-masked averages of the inputs (mean-then-encode), whereas the latter scores each soy pixel and averages the outputs (encode-then-mean). Since f_θ is nonlinear, predicting from municipal means need not match the mean of pixel-wise predictions. The slightly higher pooled R^2 on the pixel path suggests that post-training downscaling can recover municipal signal that is smoothed away when covariates are pre-averaged.

Training on the encode-then-mean path would align checkpoint selection with pixel inference and sub-municipal maps, but we kept municipal-mean training for practical reasons. Each municipality–year is one training example (municipalities times training years per LOO fold), and the full fold fits in memory. Pixel-wise training would run a forward and backward pass on every soy pixel in those municipality–years before municipal averaging, which amounts to hundreds of millions of pixel updates per epoch, rather than the tens of thousands of municipal updates per epoch. Besides the extra wall-clock time of millions of additional forward and backward passes per epoch, the data themselves become an I/O bottleneck: the network weights fit comfortably in GPU memory, as well as the tens of thousands of the mean-pixel series, but at hundreds of millions of soy pixels the time series no longer fit and must be streamed from disk in batches. Those costs made pixel-wise training too expensive relative to municipal-mean fitting, so we train mean-then-encode and report pixel-path LOO on the same checkpoints.

6.3 The 2022 drought year

The 2022 fold is the exception that matters for generalizability evaluation. It is the only drought year in the Paraná sample, so leaving it out removes every training example from that regime. The resulting negative R^2 under both evaluation paths shows that the present covariates are not enough to learn drought behaviour from wetter years alone. When 2022 remains in training and another year is held out, LOO R^2 is much more stable. This gap is mainly a data problem. If the training set includes municipalities that suffered poor yields in other years and regions, drought-like response patterns become observable rather than unique to one left-out season.[1]

Modality ablations on the same year-LOO folds show that climate carries most of the non-drought signal, Sentinel-2 less, and soil least. Omitting climate costs about 0.32 in mean pixel-path R^2 when 2022 is excluded, versus about 0.26 without Sentinel-2 and about 0.07 without soil. Soil still matters for interpretation: texture and SOC are nearly static across years, so they act as a spatial productivity prior that helps typical seasons but can worsen the drought fold. Without soil, the 2022 LOO score rises from -0.422 to -0.203, while ordinary years still favor the full three-stream model. Ranking modalities by the mean that includes 2022

[1] We have already begun that expansion by adding Rio Grande do Sul, São Paulo, and Goiás to the municipal-mean pipeline. On that multi-state year LOO setup we obtain mean R^2 about 0.46, with fold scores ranging from about 0.2 to 0.7. We do not report those numbers as main results here because the pixel-by-pixel pipeline has not yet been run at that scale, and municipal-mean scores alone would miss the downscaling objective that motivates MSTNet.

can therefore look misleading; the fair comparison for typical seasons is the mean that excludes that fold.

That same distinctiveness has the opposite effect under the spatial holdout: among the five seasons, 2022 has the highest per-year R^2 (Table 8). Since drought-year municipality–years appear on both sides of the split, the model can learn the 2022 response from the training municipalities and use it to rank held-out places within that same season. The year is then no longer an unseen regime, only unseen municipalities inside a season already represented in training.

Paraná year-LOO makes 2022 look uniquely hard only because that drought is absent from training. Once drought seasons are represented among the training years, we expect the opposite: drought years should be the easier folds, because climate and spectral inputs leave a strong imprint of stress that the model can map to yield collapse. Favourable seasons are the harder case for within-year ranking: after the seasonal level is set, the residual gap between good and poor municipalities will have more to do with management, cultivar, and planting decisions that our covariates do not observe.

6.4 Model limits

Two failure modes matter beyond the headline LOO scores. First, the model can look competent by forecasting the year well while ranking municipalities badly inside that year. Climate already explains most of the year-to-year swing on non-drought folds, so MSE rewards getting the seasonal level right; the smaller within-year gaps in PAM are easier to ignore. Second, mid-range municipalities can collapse toward a common predicted value: under plain MSE alone there is little reason to spread predictions when within-year residuals are noisy, yet absolute level must still be calibrated across years. Drought LOO stresses both failure modes at once, because the model must predict an unseen seasonal level and still order municipalities without any drought example in training. A further ceiling is the covariates themselves: management, cultivar, planting cycle, and field history are missing, and soy-masked municipal means erase within-municipality contrast that might separate productive from marginal soy inside the same PAM unit.

6.4.1 Architecture and losses

As stated above, the model might be good at forecasting the year well while ranking municipalities badly inside that year. We have a couple of ideas on how to update the architecture and losses to reward within-year ranking.

The simplest "brute-force" fix is to put the missing objective in the loss: within-year ranking or correlation terms, and a penalty when predictions inside a year are unrealistically tight. Those terms ask who is above or below that year's mean and back-propagate through both the decoder and the embedding, so the same failure mode can be fixed in either place. A year-level baseline plus a municipal deviation would separate seasonal calibration from within-year ranking more explicitly, and would make under-dispersion a measurable failure.

A generative auxiliary task might also enrich the embedding: it pressures input fidelity so trunks do not discard spectral or climate detail that a deep yield head does not need for a coarse seasonal fit. That keeps more of the input structure available for ranking, without itself teaching within-year order.

On the climate trunk specifically, Seq2Vec today is a uniform mean over valid days, unlike the NDVI-weighted pool on the spectral side; a learned or phenology-aware climate Seq2Vec that up-weights dry or hot windows could make the climate embedding more sensitive to stress without changing daily inputs or the pixel-path layout. Another way to make the embedding space richer is self-supervised pre-training of the trunks, as Xu et al. [2024] do for STNet on

large Sentinel-2 archives before downstream fine-tuning. We train MSTNet from scratch on municipal means, so having the pre-training might be helpful.

6.4.2 Features

A keen reader may notice that we did not engineer yield covariates from the streams. The spectral trunk receives Sentinel-2 band time series, the climate trunk receives daily Xavier variables, and soil enters as MapBiomas Solo attributes, without the vegetation indices, phenology summaries, or seasonal aggregates that are often used in the literature. NDVI appears only as Seq2Vec pooling weights on the spectral trunk, not as an input feature alongside the bands. This choice was conscientiously made and the working assumption is that a multi-frequency Transformer can recover the nonlinear relationships that hand-crafted indices encode. Although our expectation is that the MLP and the Transformer will recover all relevant information for the yield prediction task, in practice that recovery is not guaranteed. With only municipal PAM labels and an MSE that mostly rewards the seasonal level, the network may never be forced to form the same cues that hand-crafted features make explicit. Passing standard indices and climate stress summaries as additional channels would test whether explicit phenology and stress cues still help when the raw series are already in the model.

7 Conclusion

We presented MSTNet, a multi-frequency Transformer that learns soybean productivity from municipal PAM labels and scores soy pixels for sub-municipal maps. Sentinel-2 time series and daily Xavier climate enter parallel STNet trunks; MapBiomas Solo enters the decoder by late fusion. Training uses soy-masked municipal means for speed; after training, the same weights score every soy pixel and average to municipalities or arbitrary polygons.

Year leave-one-out on Paraná harvest years 2020–2024 is the primary score. Excluding the 2022 drought fold, municipal-mean LOO reaches mean $R^2 = 0.275$ and the pixel path reaches mean $R^2 = 0.302$ (RMSE $= 0.427$ t/ha), slightly above the municipal-mean path on the same checkpoints. Partial-season scoring approaches that full-window accuracy by February on non-drought folds, allowing an early prediction of that year's yield. Fixed-architecture modality ablations rank climate first, Sentinel-2 second, and soil last on ordinary years; soil can help typical seasons and hurt the drought fold. Under an 80/20 municipality holdout with shared seasons instead of year-level split, pooled $R^2 = 0.736$, though much of that pooled score separates years rather than places within a year.

The 2022 LOO collapse is a missing-regime problem, not evidence that drought is intrinsically hard for these covariates: when 2022 appears on both sides of a spatial split it becomes the strongest year, and we expect drought years to become easier once stress seasons are present in training, while favourable seasons remain harder for within-year ranking. Intramunicipal maps stay qualitative until field yields can score them.

Further work follows the discussion in Section 6.4. On the data side, we are still lacking broader geographic and drought coverage and pixel-path evaluation at multi-state scale, so stress regimes are represented in training rather than left as LOO surprises. On the model side, within-year ranking or correlation losses, a year-level baseline plus municipal deviation, generative auxiliary tasks, and a stress-aware climate Seq2Vec should push the stack beyond seasonal-level fit, while adding standard vegetation indices and climate stress summaries as explicit features will test whether hand-crafted cues still help when the raw series are already present.

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A Pixel-level MSTNet pipeline

Figure 1 in the main text shows the municipal-mean path used for training and for the municipal-mean LOO metrics in Section 4. Figure 7 diagrams the pixel-wise alternative used after training: soy-masked Sentinel-2 pixels are encoded one by one, climate is broadcast across pixels, soil enters by late fusion, and the mean of the pixel predictions $\hat{y}_p$ is compared with PAM (11). We do not train with that path; it is much slower than municipal-mean training while targeting the same labels. The diagram is still useful because the post-training evaluation in Section 5 follows the same forward path without backpropagating through every pixel, and the resulting field $\{\hat{y}_p\}$ can be kept as a map or averaged over any polygon.

The network weights are shared with the municipal-mean training path: the MSTNet blocks in Figure 2 are unchanged, and only the input aggregation differs, as in Figure 7.

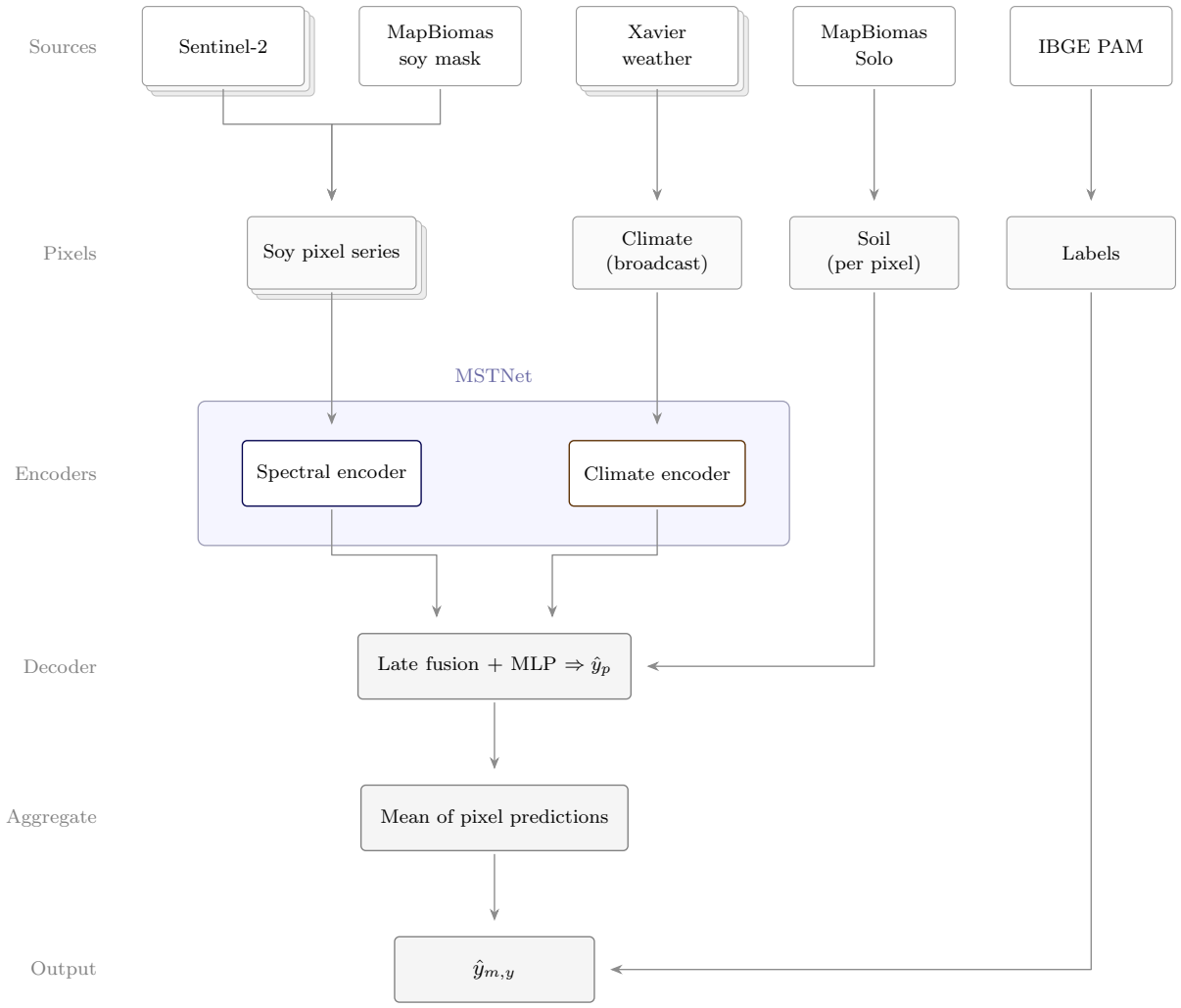

Figure 7. Pixel-wise MSTNet data pipeline used for post-training evaluation and sub-municipal maps, not for training. Soy-masked Sentinel-2 pixels are encoded individually; municipal Xavier climate is broadcast to pixels and soil enters by late fusion. The mean of pixel predictions $\hat{y}_p$ yields $\hat{y}_{m,y}$ for comparison with IBGE PAM.

B Municipal-mean LOO diagnostics

Section 4.3 reports full-season municipal-mean LOO metrics. Here we add the municipal-mean counterparts of the partial-season curves, scatterplots, Paraná error maps, and modality ablations from Section 5. There is no Mamboré panel for this path because municipal-mean inputs do not produce intramunicipal scores.

B.1 Partial-season evaluation

Table 6 and Figure 8 report municipal-mean R^2 when only the first k season months are kept. As on the pixel path, early truncations are strongly negative; excluding 2022, mean R^2 reaches -0.034, 0.269, and 0.275 at four, five, and six months.

B.2 Scatterplots and spatial maps

Figure 9 shows predicted versus observed municipal productivity under the municipal-mean path for every LOO year. Figure 10 shows municipal error maps for the best and worst LOO years; Figure 11 shows 2020, 2023, and 2024.

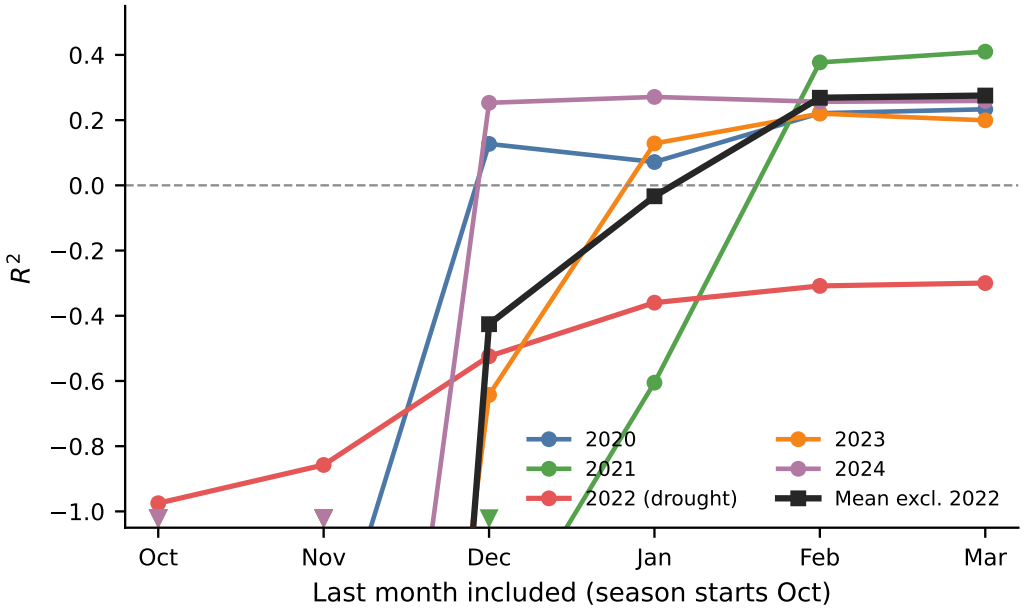

Figure 8. Municipal-mean LOO R^2 versus season length. Each curve is one left-out year; the black squares are the mean over non-drought folds. Points below the axis floor are clipped and marked with downward triangles.

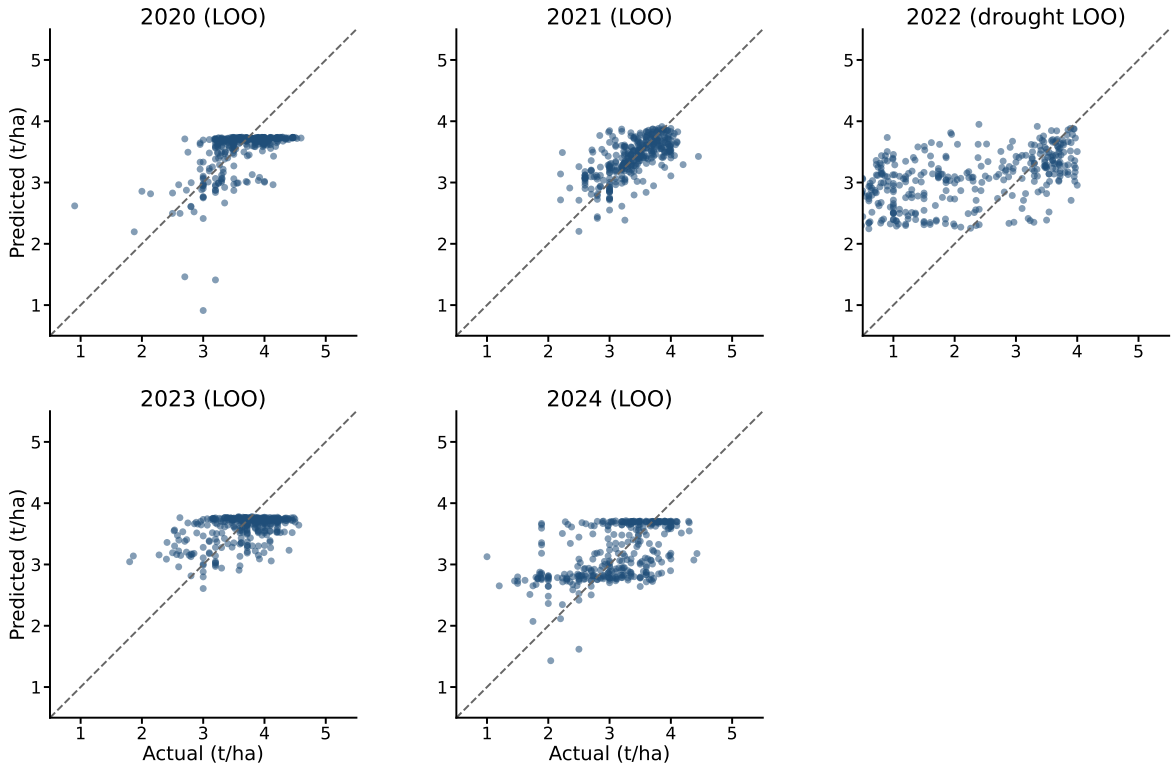

Figure 9. Predicted versus observed municipal soybean productivity (t/ha) **for each year-LOO fold under the municipal-mean path.** Every panel evaluates only the left-out year. The dashed line is identity.

Table 6. Municipal-mean year-LOO R^2 under incomplete seasons. Each column keeps only the first k months of the Oct–Mar window ($k = 6$ matches Table 2).

LOO year	Oct	Nov	Dec	Jan	Feb	Mar
2020	−19.078	−1.517	0.127	0.072	0.221	0.233
2021	−19.950	−18.097	−1.441	−0.605	0.377	0.410
2022	−0.975	−0.858	−0.524	−0.360	−0.308	−0.300
2023	−3.056	−4.617	−0.643	0.128	0.220	0.199
2024	−4.293	−3.539	0.253	0.271	0.256	0.259
Mean	−9.470	−5.725	−0.446	−0.099	0.153	0.160
Mean excl. 2022	−11.594	−6.942	−0.426	−0.034	0.269	0.275

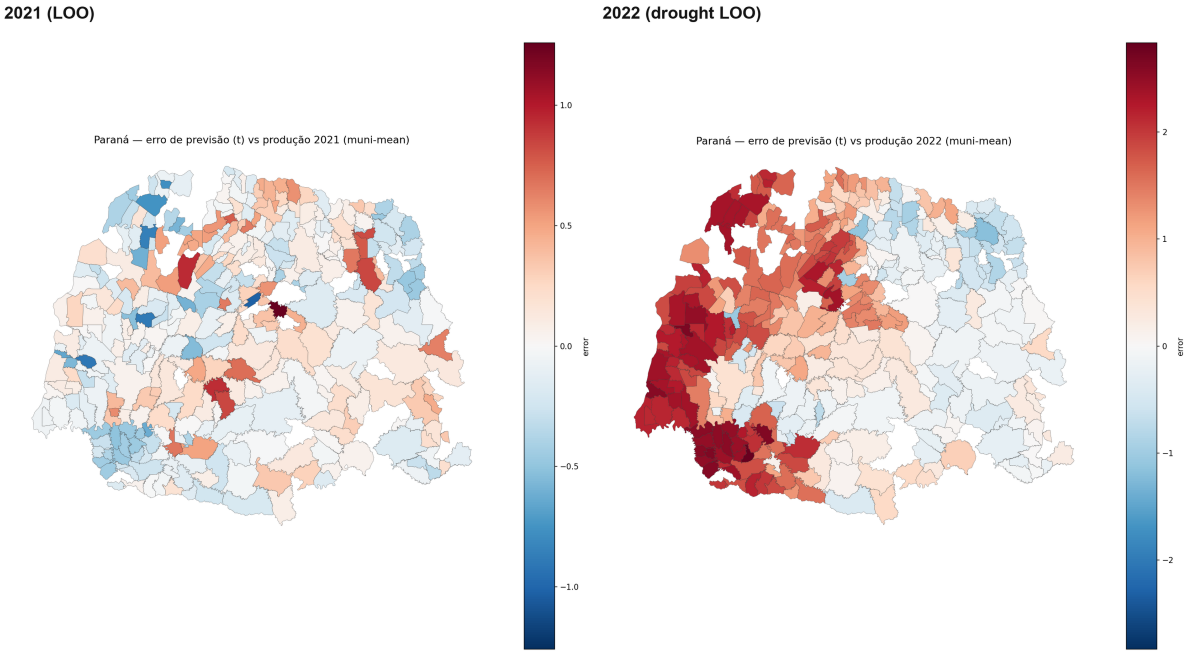

Figure 10. Municipal aggregate prediction error maps under the municipal-mean path for LOO years 2021 and 2022.

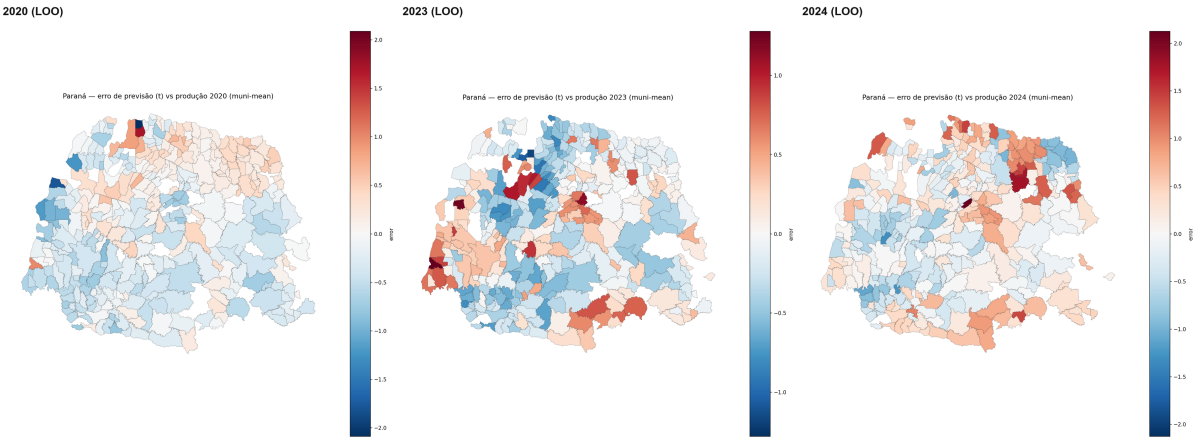

Figure 11. Municipal aggregate prediction error maps under the municipal-mean path for LOO years 2020, 2023, and 2024.

B.3 Modality ablations

Table 7 reports the same leave-one-modality-out runs as Table 5, scored on the municipal-mean path used in training. Excluding 2022, full MSTNet has mean $R^2 = 0.275$. Omitting Sentinel-2 costs about 0.24 in mean R^2, omitting climate about 0.15, and omitting soil about 0.03.

Table 7. Year-LOO municipal-mean metrics for full MSTNet and leave-one-modality-out ablations. Each ablation uses the same hyperparameters as full MSTNet and drops only one input stream. RMSE is in t/ha. Full MSTNet keeps S2+climate+soil; No soil keeps S2+climate; No climate keeps S2+soil; No Sentinel-2 keeps climate+soil.

Variant	Metric	2020	2021	2022	2023	2024	Mean	Excl. 2022
Full MSTNet	R^2	0.233	0.410	−0.300	0.199	0.259	0.160	0.275
	RMSE	0.411	0.320	1.342	0.433	0.577	0.617	0.435
No soil	R^2	0.205	0.343	−0.159	0.163	0.290	0.169	0.250
	RMSE	0.419	0.338	1.267	0.442	0.565	0.606	0.441
No climate	R^2	0.257	0.203	−0.290	0.158	−0.130	0.040	0.122
	RMSE	0.405	0.372	1.337	0.444	0.713	0.654	0.483
No Sentinel-2	R^2	−0.434	0.187	−0.302	0.095	0.299	−0.031	0.036
	RMSE	0.562	0.376	1.343	0.460	0.562	0.661	0.490

C 80/20 municipality holdout

Section 6.1 contrasts year-LOO with the municipality holdout used by Song et al. [2022]: randomly hold out 20% of municipalities for testing, keep the other 80% for training, and include every harvest year in both pools. Table 8 reports that holdout for the same Paraná MSTNet architecture as Table 2 (one Transformer layer, $d = 128$, four heads, $d_{\text{inner}} = 64$, dropout 0.4), trained once on the municipal-mean path with the 80% training municipalities and scored on the held-out 20%. Per-year rows restrict the holdout to a single harvest year; Mean averages those five yearly scores; Pooled scores all holdout municipality–years together. Pooled $R^2 = 0.736$ (RMSE 0.448 t/ha) is much higher than year-LOO on the same architecture, while the yearly R^2 values (0.283–0.636) show that much of the pooled score comes from shared seasonal structure across years rather than pure spatial interpolation within a fixed season.

Table 8. 80/20 municipality holdout for Paraná MSTNet under the municipal-mean path. Twenty percent of municipalities are held out for validation and test; all harvest years 2020–2024 appear in both pools. RMSE and MAE are in t/ha. Mean is the unweighted average of the five yearly scores; Pooled aggregates all holdout municipality–years.

Subset	n	RMSE	MAE	R^2
2020	70	0.330	0.262	0.426
2021	68	0.305	0.233	0.409
2022	69	0.694	0.468	0.636
2023	68	0.411	0.323	0.283
2024	68	0.385	0.300	0.581
Mean	—	0.425	0.317	0.467
Pooled	343	0.448	0.317	0.736

D Additional pixel-path heatmaps

Section 5 highlights the best and worst LOO years for the pixel-path Paraná and Mamboré maps. Figures 12 and 13 report the remaining folds, 2020, 2023, and 2024. Figure 14 stacks the 2022 Mamboré productivity map with mean NDVI and peak NDVI on the same soy pixels; we use 2022 because its peak NDVI field is less homogeneous than in the other LOO years, so intramunicipal contrast is easier to see. That comparison is qualitative only.

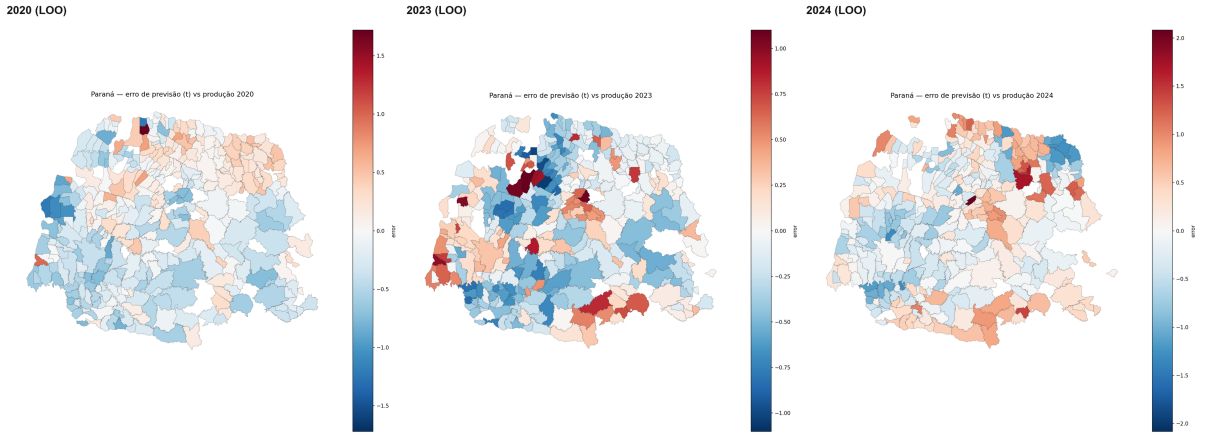

Figure 12. Municipal aggregate prediction error maps under the pixel path for LOO years 2020, 2023, and 2024.

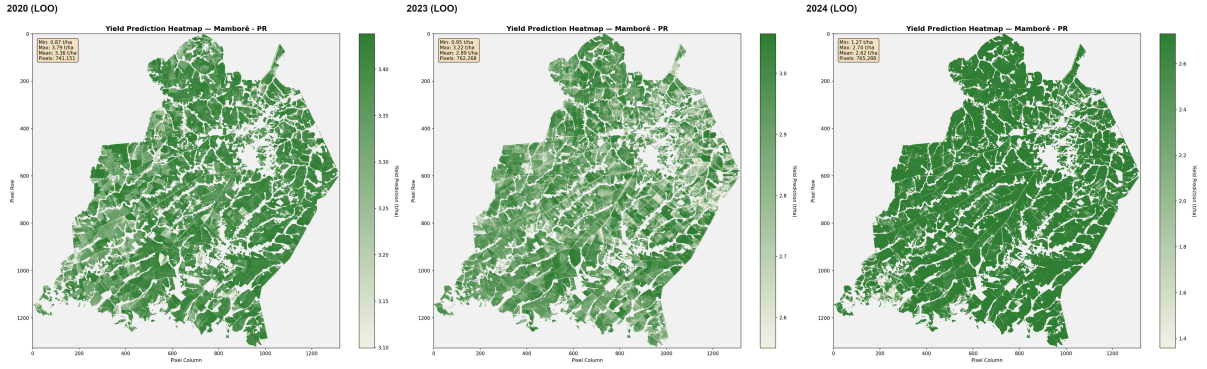

Figure 13. Intramunicipal Mamboré productivity maps under the pixel path for LOO years 2020, 2023, and 2024.

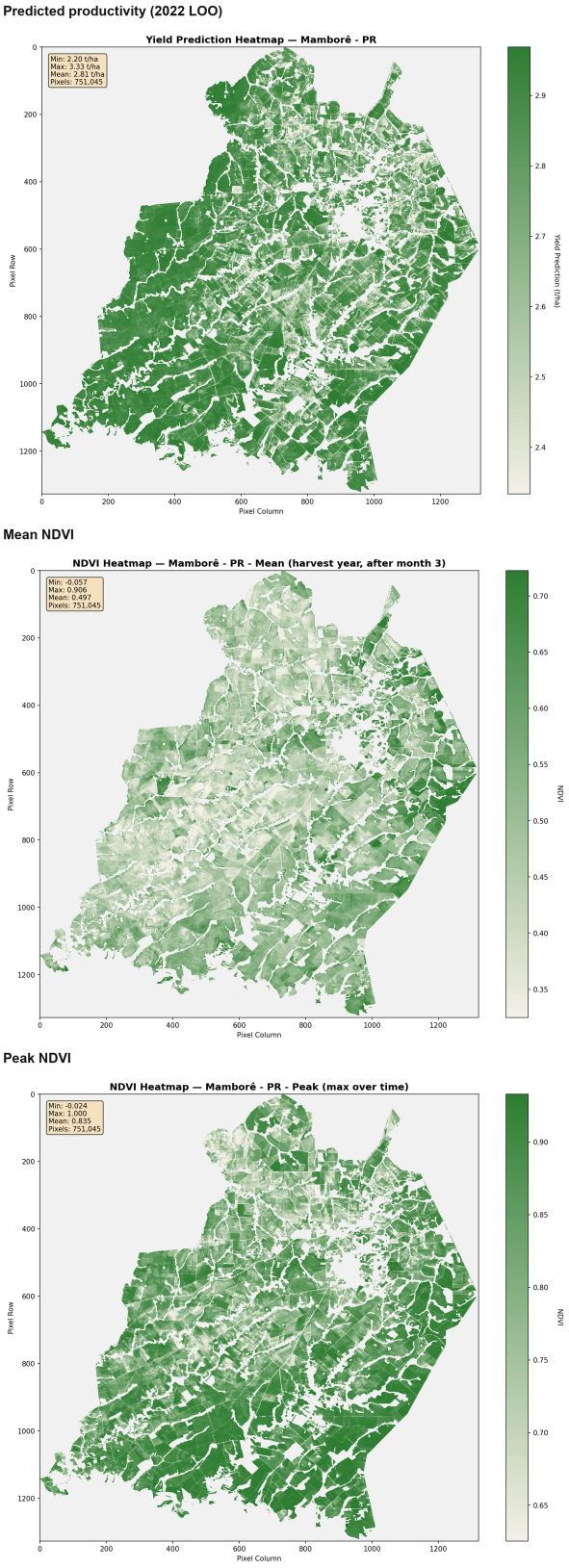

Figure 14. Mamboré under the 2022 LOO checkpoint: predicted productivity, mean NDVI, and peak NDVI on soy pixels, stacked so that spatial patterns can be compared directly.